Supplement to: How BGP Origin Changes Spread and Return, and How Well Early Observations Forecast Route Leaks

Anonymous Author(s)

Abstract

This supplement holds the revision analyses, further tables and the full R2 protocol record of the paper. References without the prefix S point to the paper, for example Section 4 or Appendix D.

S1 Revision analyses

S1.1 Month boundary

First-of-month identifier dates hold 15 % of R1 and 13 % of R2 events, against 3.3 % and 4.4 % for uniform dates (Section 8). In R1, 65 of the 99 volume-selected events fall on that day, and the R2 scores lack the selection reason.

Without the leaks of the first day, C2 moves toward SAGE with history in both periods and at both horizons (Table S5). Without days 1–3, R2 moves the other way. In R1 at 30 s, C2 becomes +0.0066 [+0.0008, +0.0118], and in R2 we still detect no difference. The advantage of online history on its native candidates shrinks by 16–25 % without day 1 and by 8–20 % without days 1–3. This descriptive revision analysis lies outside the prespecified families, and its intervals are unadjusted.

S1.2 Overview of the revision analyses

Seven further revision analyses reuse the per-event scores of R1 or R2, besides the month-boundary and recovery day-block analyses (Tables S4 and S5). We designed all of them after the R1 and R2 results (Table S2). They read no protected label, change no published estimate and form no test family, and their intervals are unadjusted. The first redraws the intervals of C2 and of the native contrast with the leaker AS as a resampling unit (Table S6). A second variant draws leakers and days independently and weights each event by the product of their multiplicities.

The second analysis applies both variants to the month-boundary subsets (Table S9). Every interval of the native contrast excludes zero, in both periods, at both horizons and in every subset. In R1 at 30 s without first-of-month leaks, C2 is +0.0066, with the intervals [+0.0008, +0.0118] over day blocks and [+0.0007, +0.0120] over leakers. If we draw leakers and days together, its interval is [−0.0005, +0.0136] and includes zero. Without days 1–3, the same holds, with [−0.0011, +0.0147]. At 60 s in R1, and in every R2 cell, every interval of C2 includes zero.

The third analysis counts the ASes that each set forecast adds to the seeds, over all scored events (Table S7). The R2 records hold the Jaccard of each forecast but not the forecast set. Two identities still give R2 values (Table S7). The two-hour target holds the seeds, so persistence reaches the Jaccard $s/(s+R)$ for s seeds and R new members. On an event without new members, every addition is false, so a forecast with Jaccard J adds $s(1/J-1)$ ASes. Both identities hold on every R1 record.

On an event with new members, the Jaccard does not separate true from false additions. R2 therefore has no counts of added ASes on these events and no share of forecasts that add nothing. The share of events with the Jaccard of persistence is an upper limit of this share. For SAGE with history in R2 it is 0.937 at 30 s and 0.972 at 60 s. In R1, this limit equals the recorded share at both horizons, 0.946 and 0.996.

In R2, persistence reaches a Jaccard of 0.671 at 30 s and 0.776 at 60 s. At 30 s, SAGE with history differs from it by −0.000210 [−0.000482, +0.000018]. At 60 s, the difference is −0.000082 [−0.000187, −0.000012], and this interval excludes zero. Online history with the mean size differs from persistence by −0.2900 [−0.3154, −0.2627] and −0.2919 [−0.3156, −0.2675]. At 30 s, it makes false additions on all 285 events without new members, 52.56 per event, and with the median size on 78.6 % of them. SAGE with history makes false additions on 1.05 % of these events.

Events without new members carry 55 % of the Jaccard gap between persistence and online history with the mean size at 30 s, and 67 % at 60 s. In R1, they carry 54 % and 69 %. R2 thus repeats the set errors of R1.

Table S7 also gives the absolute P@10 of each arm, and its differences reproduce every published P@10 contrast. At 30 s, SAGE with history and online history on the shared and on its native candidates reach 0.271, 0.270 and 0.270 in R1. In R2, they reach 0.271, 0.295 and 0.291. Native minus shared history is −0.0002 [−0.0013, +0.0011] in R1 and −0.0041 [−0.0070, −0.0012] in R2. At 60 s, it is −0.0002 [−0.0015, +0.0011] and −0.0043 [−0.0078, −0.0008]. The native candidates therefore add no precision at ten, although they add AP.

The per-event score files hold no identities of new members or of forecast ASes. Of the member ASes, the R2 files name only the seeds, and the R1 files name none. From these files we therefore cannot split new members into ASes that recur from earlier events and other ASes. Such a split would also be almost empty on one side. Native candidates cover 99.96 % and 99.94 % of the new members of an R1 event on average.

S1.3 Leaker recurrence

The fourth analysis classes each test event by its leaker, the AS that the event identifier names. For all 3,543 R1 events, this AS matches the event table. A *repeat* leaker is the leaker of at least one completed earlier event of the history library, and any other leaker is *first-seen*. An earlier event is complete if its onset lies at least 2 h before this onset.

The library is the one that online history reads, the successful leak events of the selected release tables plus the earlier R2 test events. For R1 these tables cover 2023–2025, and for R2 they cover 2023 to 31 August 2026. The other 22 identified R2 events exist only in the custodian table. A leaker whose only earlier event is one of them therefore counts as first-seen. A repeat leaker is thus an AS-level notion relative to these selections, not to all leaks. A persistent valley path from a complex AS relationship would also recur under the same leaker.

Repeat leakers hold 84 % of the eligible R1 events at 30 s (2,241 of 2,661) and 88 % of the eligible R2 events (880 of 1,004). Of the eligible R1 events, 420 have no earlier event of their leaker, 282 have one and 633 have two to four. Another 1,003 have five to 19, and 323 have 20 or more.

In R1 at 30 s, C2 is +0.0131 [+0.0058, +0.0204] on first-seen and +0.0040 [−0.0018, +0.0094] on repeat leakers (Table S8). On first-seen leakers, the interval still excludes zero if we draw leakers and days together, with [+0.0021, +0.0247]. The native contrast is −0.0112 on first-seen and −0.0260 on repeat leakers. Per event, the lead of SAGE with history is larger on first-seen leakers, but repeat leakers carry 62 % of its mean over all events. By point estimate, the native advantage of history is larger on repeat leakers. With day blocks, the difference between the classes excludes zero for both contrasts, but with leakers and days drawn together it includes zero.

At 60 s in R1, the estimates point the same way, but the first-seen interval of C2 includes zero if we draw leakers and days together. R2 has only 124 first-seen events at 30 s. Its estimates also point the same way, with C2 at +0.0087 and −0.0009 and the native contrast at −0.0199 and −0.0312. Every R2 interval of C2 and every R2 difference between the classes includes zero. If the library holds only events before the test period, repeat leakers hold 77 % of the eligible R1 events. C2 is then +0.0122 on first-seen and +0.0034 on repeat leakers.

S1.4 Same-leaker baseline (post hoc)

We designed this baseline after the R1 results were known, so it is post hoc and descriptive and lies outside the prespecified families of R1. The *same-leaker-only ranking* orders the native candidates of online history by their participation in completed earlier events of the same leaker. The *same-leaker-first ranking* breaks the ties of this count by the participation in all completed earlier events. On first-seen leakers it therefore equals online history. For R1, the released 2025 event table gives the seeds and the targets, which are measurement data, not protected labels.

We did not test either ranking in R2. The R2 protocol did not declare them, so a further label read for them would make R2 post hoc. The re-scorings scored only declared lanes.

The released seed sets count an AS from its first observation, and the forecast input counts it from its availability. On some events, the released sets therefore hold more seeds, up to 55 at 30 s, and the files do not show which ones. All three rankings use the released seed sets, on the 2,635 of 2,661 eligible events that keep a new member under them. On 1,866 events the two seed sets are equal, and there our reconstruction reproduces every published record of online history to 10^{-12}. Only these exact-input events allow a comparison with SAGE with history. They are a selected subset, and online history reaches 0.204 AP on them, against 0.236 on all eligible events.

At 30 s, the same-leaker-only ranking differs from online history by +0.0019 [−0.0115, +0.0140] AP (Table S8). It gains +0.0408 [+0.0277, +0.0530] on repeat leakers and is 0.2044 lower on first-seen leakers, which have no such events. The same-leaker-first ranking gains +0.0729 [+0.0643, +0.0812] AP over online history. If we draw leakers and days together, its interval is [+0.0589, +0.0880]. At 60 s, this gain is +0.0610 [+0.0516, +0.0706].

Unlike the native candidates, it also gains in precision at ten, with +0.0475 [+0.0386, +0.0556] in P@10 at 30 s. On the exact-input events, it reaches +0.1043 [+0.0933, +0.1151] AP over SAGE with history, and online history +0.0214 [+0.0144, +0.0288]. This compares complete systems, because SAGE ranks the shared candidates, whose identities the per-event records do not hold.

S1.5 Issue-time slice

The fifth analysis compares the set forecasts on the issue-time slice with persistence (Table S10). The slice and its threshold are prespecified, but this contrast is not.

S1.6 Leak audit

The sixth revision analysis audits 30 R1 test leaks in each of four strata: Radar or volume, and day 1 or not (Table S11). A script applies fixed rules in a fixed order, without human labels and without the arm results. A leak is *ambiguous* if we recover no violating triple, or if Radar matched it but no triple still violates in the next month. It is also ambiguous if its route was visible from the start of the look-back and its violating triples were new the month before. Otherwise, a leak whose triples no longer violate the next month has a *changed or missing* relationship, through a changed type or a missing adjacency.

A leak is *relationship-consistent* if a triple violates valley-freeness under the CAIDA relationships of its month and the next month, and no ambiguity rule applies. A sibling, partial transit or hybrid link that CAIDA infers wrongly in both months also passes. For example, the class holds a route that AS3356 exports to AS209, which both belong to Lumen, and five events of Hurricane Electric (AS6939). The routes of these five events were visible from the start of the look-back, and Radar matched none of them. By the independent evidence of this section, all six are uncertain.

The identifier keeps triples whose first and last AS are the same, such as 131111-131736-131111. They are 4 of the 16,602 audited triples, and no class rests on them alone. The identifier also applies the CAIDA relationships, which CAIDA infers from IPv4 paths, to IPv6 paths, and 21 audited leaks are IPv6-only. This can explain part of the two IPv6-only events of Hurricane Electric in the relationship-consistent class. Under CAIDA's IPv6 relationships, 15 to 30 % of their triples still violate valley-freeness in both months, so both events stay in the class.

We check siblings with the CAIDA AS-to-organisation snapshot of each leak's month, which CAIDA infers from WHOIS records, not from routes. In 96 of the 97 relationship-consistent leaks, a triple that violates in both months has no sibling pair of adjacent ASes.

In the other leak and in three ambiguous leaks, every such triple has a sibling pair inside the US Department of Defense. All four are volume-selected leaks of day 1, and no Radar-matched leak has a sibling pair. A sibling pair suggests a route inside one organisation, so these four events are likely not leaks. The absence of a pair does not confirm a leak, and the data miss some siblings, such as AS3356 and AS209. This check therefore does not measure precision.

A second class rates the same leaks by fixed rules, without the identifier's relationship test and without the arm results. Two addenda, for missing data and the two-hour end state, followed the first RIPEstat downloads, preceded any contrast and can only remove support. No support rests on a split window.

A leak is independently supported if a Radar label names its leaker and the two neighbours of one of its triples, in either order. It is also supported if a new route in RIS [4] ends within two hours. It is independently contradicted if every triple joins two ASes of one organisation, and uncertain otherwise. IPv6-only leaks without a Radar label are uncertain, because the route check uses IPv4 triples only.

Of the 120 leaks, 71 are independently supported, 50 by a Radar label and 21 by a short new route, 4 contradicted and 45 uncertain. Of the 71, 31 use no relationship input: the 21 short new routes and 10 Radar-supported leaks that also pass the route check. A short new route fits a leak, but also path exploration or a short traffic-engineering change. Support is therefore not proof of a leak, and these classes corroborate and do not validate.

On the 61 eligible supported leaks, C2 is +0.025 [+0.005, +0.056] and the native contrast −0.003 [−0.030, +0.030], in AP at 30 s. This unweighted subset is not a cohort estimate, and both intervals contain the values over all eligible events. For 25 of the 50 leaks that Radar supports, RIS or the previous month already shows the path before the identifier time. For these, the class supports the leak, not its identified start. This subset holds only 1 of the 30 sampled volume-selected leaks of day 1, so it cannot show why the native contrast differs. Radar also uses CAIDA relationships, among others, and the same collectors, so this class corroborates leaks but does not measure precision.

On the volume-selected leaks of day 1, the native contrast is −0.151 AP at 30 s, against −0.024 and −0.019 on the Radar-matched strata. These 65 events carry 16 % of the native advantage of history over all 2,661 eligible events. On these Radar-matched strata, C2 is +0.0009 and +0.0064, against +0.0054 on all eligible events. Without the volume-selected events, C2 is +0.0056 (95 % interval −0.0001 to +0.0109) and native is −0.0202 (−0.0264 to −0.0144).

S1.7 Native and shared history

The seventh revision analysis gives the paired interval of native minus shared online history, with the statistic of the month-boundary analysis. Native history gains +0.029 [+0.026, +0.032] AP in R1 and +0.030 [+0.025, +0.035] in R2 at 30 s (Section 7.2).

S2 Further methods and results

Table S1 gives the descriptive comparison of the two size estimates of history.

S2.1 R2 AP by number of new members

In R2 at 30 s, SAGE with history and shared online history reach a mean AP of 0.215, and online history on native candidates 0.245. In strata without tests, SAGE with history

Table S1: Online history as a set forecast, with the mean or the median final size of its 20 nearest past events, and SAGE with history. Jaccard against the full target over all events, and paired differences with 95 % intervals over seeds and day blocks. SAGE: online history (mean) plus C2 of Table 8, mean over five seeds. Descriptive, no test family.

		Jaccard				
	Δ (s)	mean	median	SAGE	median minus mean	SAGE with history minus median history
R1	30	0.384	0.586	0.702	+0.203 [+0.193, +0.211]	+0.116 [+0.109, +0.123]
R1	60	0.477	0.728	0.801	+0.251 [+0.240, +0.261]	+0.072 [+0.066, +0.078]
R2	30	0.381	0.569	0.671	+0.188 [+0.172, +0.203]	+0.102 [+0.089, +0.115]
R2	60	0.484	0.706	0.776	+0.223 [+0.206, +0.240]	+0.069 [+0.059, +0.079]

reaches 0.07 on events with 1–9 new members (37 % of eligible events), against 0.04–0.05 for online history. On events with 100 or more, it reaches 0.55, against 0.58 for online history on the same candidates and 0.68 on its native candidates. The mean of C2 thus combines a lead of SAGE with history on small events and a lead of online history on large ones.

S3 Protocol record

S3.1 Choices that affect reported numbers

- *Primary metric.* We changed the primary metric from Jaccard to AP over new members after the development fits. Under full-target Jaccard with one global threshold, every learned arm stayed within 0.0002 Jaccard of persistence on the broad development cohort. A development diagnosis had already shown pooled ranking results on the validation split, so this change was not blind to rankings. We fixed the estimand and its rule for tied scores before any per-event ranking number of R1 or R2 existed. We also fixed the cut of precision at ten before these numbers existed.
- *Temporal models.* TGN and DyRep-style select their configuration by validation Jaccard, which can disadvantage them under AP.
- *Frozen model at 30 s.* One frozen R1 model, at 30 s and training seed 0, cannot run under the frozen software, and the other four seeds can. The evaluator needs the same seeds for all arms at a horizon, so we decided to evaluate the frozen model at 60 s only. We sealed an amended retention record with 345 runnable roles and 55 retained failed roles, out of 400 declared roles. C5 and C6 at 30 s therefore stay in the Holm family with $p = 1$ and no estimate.
- *Headline contrast and wording.* We named C2 at 30 s the headline contrast at about 21:00 UTC on 29 October 2026. We committed the outcome wording at 17:17 UTC on the same day. Both happened before the primary evaluator joined labels to forecasts. The protocol had already named C2 and C5 as its open question.
- *Lane inputs.* The stable-peer lane keeps the original observed membership in its history library. Its inputs and targets use repaired sources for 1,010 events of

Table S2: Chronology of R2 (UTC). On 17 and 18 October, the times are the commit times of the records.

16 Oct 2026	Evaluation window closes (exclusive end).
17 Oct	Protocol written, with full-target Jaccard as primary metric, before any R2 label exists.
17 Oct, 18:34	Primary metric changed to AP over new members.
17 Oct, 21:17	Merge code of the custodian committed.
18 Oct, 06:32	Extension rule applied to validation predictions only.
18 Oct, 06:38	Custodian labels complete. Start time of their computation not recorded.
28 Oct	Freeze of models, thresholds, code and inputs.
29 Oct, 00:20	Amended freeze: frozen R1 model at 60 s only.
29 Oct, 05:22	All forecasts written and hashed; seal written.
29 Oct, 08:06	Label file for the evaluator built, after four failed runs that read label records.
29 Oct, 17:17	Outcome wording committed.
29 Oct, 21:00	C2 at 30 s named the headline contrast.
30 Oct, 08:53	Labels joined to forecasts once; primary result written (attempt 12).
30 Oct, 11:28	One disclosed re-scoring of the secondary lanes approved.
30 Oct–1 Nov	Revision analyses of the R1 and R2 scores committed from 30 Oct, 16:11, to 1 Nov, 02:28, after the primary result.
31 Oct, 08:46	Second label read, for the secondary lanes, with the primary result recomputed as a check.
31 Oct, 21:53	Re-scoring run ends, with the stable-peer lane failed.
31 Oct, 23:24	A further re-scoring of the stable-peer lane approved.
1 Nov, 02:59	Third label read, for the stable-peer lane only, with the primary result recomputed as a check.
1 Nov, 03:29	Stable-lane re-scoring run ends.
1 Nov, 06:06	Last revision analyses committed (R1 ranking file, leak-audit evidence); they read no R2 label.

2026 and recovered collector data for 1,000 more. For the reduced-collector lane, the coordinating session derived the partial-input flag of each event as the union over its two panels.

S3.2 Access to protected data before the evaluation

Four failed runs of the label-preparation step read label records before the evaluation, and nobody inspected an outcome. About 19 % of the lines of the access log of the custodian are corrupt, so the log cannot show that no other read took place. The freeze of 28 October came ten days after the custodian computed the labels.

By design, online history and the history feature read the matured targets of earlier evaluation-period events through the custodian (Table 7). "Joined once" refers to the scoring only. Earlier files hold baseline results with a May 2026 split, so July and August 2026 had been scored before. We did not establish whether these results informed a decision.

Our 2026 replay before R2 covers onsets from 1 January to 31 August 2026, and its windows end three hours after the onset. For the last two leaks of August, it therefore read archives of 1 September up to 02:07 UTC and kept only their anomalous routes. The records name no later input of the baseline run.

S3.3 Lane definitions

Candidate expansion (H1) adds to the candidates of each event the 500 ASNs with most participation in completed earlier events. Development validation Jaccard chose this number among 100, 500 and 1,000. The stable-peer lane keeps only the messages of the 639 peer identities that all 44 monthly peer lists of the ten collectors hold. We chose these peers from coverage data before any outcome, and the lane rebuilds the seeds, the inputs and the two-hour target from their messages. The frozen models score it without retraining, and an event without stable-peer input is an abstention with an empty forecast.

S3.4 Re-scorings of the secondary lanes

On 30 October, the authors approved one disclosed re-scoring of the secondary lanes, with the same float32 comparison. It read the labels a second time, on 31 October at 08:46 UTC. It also recomputed the primary result as a check, which matched attempt 12 apart from the path of its marker file. The primary result stays the one of attempt 12. Its stable-peer lane failed at an event whose stable-peer target and forecast were both empty, where the Jaccard is undefined. The run went on and scored the other families, 266 of the 344 secondary roles.

The completion gate of this run therefore refused it, so the run is not certified. Its record also marks the outcome as not authorised for release, which each driver records by design. The data validator that we approved for this run returns FAIL on five of its 54 checks. Four of them follow from the stable-lane failure. The fifth expects the baseline preparation from the checkpoints of attempt 12, but a reviewed fix prepared the baseline again without labels. Of the other 49 checks, 48 pass, including the identity of the primary result, and one only reports the size of the access log.

The authors then approved a second re-scoring, of the stable-peer lane only, with one added rule. An event whose stable-peer target and forecast are both empty is not scorable in this lane. The rule removes such events from every metric of the lane and counts them. Only the 29 events at 30 s and the 25 at 60 s without input have an empty forecast. These sets are the same in every arm, so the arms stay paired. This re-scoring read the labels a third time, on 1 November at 02:59 UTC.

It removed the same 3 events at both horizons, scored all 24 roles, and recomputed the primary result identically apart from its marker path. Its completion gate passed, with the other eight families recorded as out of scope. We wrote no separate data validator for it, and an independent review checked its record and the extraction of Table S12. Thus the scored lanes come from two runs, and no single check certifies their union. The printed lane values are partial verified results, not a certified completion of the evaluation. The reviews approved each driver before it ran, and the validator and the gates check what the runs wrote.

Seven attempts stopped at checks that refused an input. The causes were an inventory limit of one million files, a markdown file read as JSON, and review paths recorded on another machine. The others were a reviewer name, one omission in our evidence builder, and the float32 check of Section S3.5. We stopped the other four ourselves. Three were too slow or did not fit in memory. The fourth needed a correction that an offline label-free check had found.

S3.5 In-process corrections of the evaluator

We changed no file of the evaluator. We wrapped its functions in the running process, and the ledger records every wrapper with its hash. None of them changes a forecast. Apart from the opening wrapper of each disclosed re-scoring, none of them reads the label file again. The parallel scoring adapters and the float32 version of the membership scorer receive the labels that the evaluator loaded, as the pinned scorers do. Only the not-scorable rule changes a scored population with labels, and it uses only whether a stable-peer target is empty.

- *Speed.* The origin lane prepares in parallel, and a wrapper caches input projections. A fast parser reads

large JSON files, and a check of file times replaces a serial re-hash of the inventory.

- *Preparation checkpoints.* A wrapper stores label-free preparation results and reuses them after a refusal.
- *Inventory.* The inventory drops the custody inputs of the origin construction, which the evaluator does not read. This keeps it under its limit of one million files.
- *Decision check in float32.* The evaluator recomputes each committed decision as $p \geq$ threshold in float64, but the models compare in float32. One event of one arm has p equal to float32(0.94) at threshold 0.94, which the two precisions decide differently. The check now compares in float32, and it then matches every committed decision of all 281 roles.
- *Stable lane.* The persistence check of the stable lane accepts rows without a candidate list. These are exactly the 29 and 25 events with unavailable inputs.
- *Second and third opening.* The evaluator refuses a second label opening when the marker of the first one exists. For each re-scoring, a wrapper requires exactly the markers of all earlier openings. It gives the evaluator a new marker path in the new records directory. The evaluator then writes the new marker right before its single label read.
- *Parallel scoring.* In the re-scorings, adapters split the per-event scoring of large cells over eight workers, with the unchanged scorers.
- *Not-scorable rule.* In the stable-lane re-scoring only, a wrapper removes the events whose stable-peer target and forecast are both empty before the unchanged scorer runs.

S3.6 AI-assisted review of the evaluation steps

AI-agent sessions also reviewed the steps that touched protected evaluation data (Section 7.3) and drafts of this paper. Until 27 October 2026, the reviewing session was never the session that prepared the step, with one recorded exception. From 27 October, one coordinating AI session approved every step of the sealed evaluation after a short check (Section S3.1). The check was a skim of a short summary of each step: what changed, the tests and their output, and the risks. It also took the scientific decisions of that phase together with the authors. The authors directed every session, so these reviews are not independent of the authors.

S4 Release details

This section gives the full release text of Appendix B.

The 16 October package. The package holds the replay code and the event identifiers with the observed path-AS sets for January 2022 to August 2026. It flags unavailable source windows and unsupported withdrawal baselines. It also holds the RPKI validity tables, the recovery pairs, the beacon measurements, and the retained checkpoints and per-event evaluation tables of the slow-regime experiments. It gives instructions to reproduce the main tables and figures with Python 3.12.

The package dates from 16 October 2026, before R1 and R2, so it holds no R1 or R2 results, ledgers or review records. Its manifest lists the size and SHA-256 hash of every file, so a reader can check any copy against it. The manifest itself has the SHA-256 hash `a05edbba0c4fb182b55f3019a35682986dc2d27530daeb70ef54030a86b6596b`.

The scoring bundle. On 1 November 2026, we added a scoring bundle in the directory `scoring-bundle-r1r2` of the same repository, with these contents:

- the R1 report, the R1 per-event score files of every arm, the R2 primary result and the R2 per-event scores
- the R1 target sets as scored, in the ranking file, and the R2 label file, both derived from public archives
- the seal of the R2 forecasts, with the hash of the forecasts of each arm, and the ledger of the R1 forecasts
- a ranking file of the R1 forecasts, with its checker
- the evaluator, the in-process wrappers of Section S3.5 and their hashes
- the freeze, seal and deviation ledgers, the access log of the custodian and the review records
- the scripts, inputs and outputs of the revision analyses
- the records of the two re-scorings of the secondary lanes, with their completion and validator records, the validator and the scripts that extract their table
- the inputs of the leak audit, except the CAIDA files, graphs built from them and RIPEstat downloads, which a reader obtains or rebuilds
- a manifest with the size and SHA-256 hash of every file

The per-event score files name no forecast AS and no new member, and only the R2 files name the seeds (Section S1.2). The native candidates of online history are the ASes of completed earlier leaks, so a reader can rebuild them from the released event tables. The released copies of generated files replace host names and local paths with placeholders. The bundle manifest has the SHA-256 hash `754e7a0c49365947182b34be5bb2b95ef04b71819d25e7a82fd6438bbb009b44`, and a restore script checks every file against it.

We do not release the raw R1 forecasts or the sealed R2 forecasts, which take 432 and 425 GB. Instead, a ranking file of 121 MB holds, for each R1 cell and event, the ranking positions of the new members with their ties. The file also holds,

per event, the target and input-seen seed sets as scored. From these, a checker (`r1_rank_sidecar_check.py`) recomputes AP, precision and recall at k and coverage, and matches every released value. It covers 1,008 of the 1,032 R1 cells with ranking metrics, because the forecasts of 24 history baselines no longer exist. The checker needs about 3 minutes on a workstation in its fast mode.

The revision analyses of the R1 and R2 scores run as Python scripts on a workstation in minutes, from the released per-event scores. The allowlist and trust-anchor checks of Section 3 read intermediate replay outputs and the ROA archive, which the bundle does not hold.

The released R2 per-event scores support every R2 aggregate of the paper, but re-scoring R2 needs the unreleased sealed forecasts. With them, the runner of the evaluator (`runner.py`) reads the freeze, the seal and the label file. Each of our runs requested about 2 TB of memory on one node. Attempt 12 ran for 8 hours, and the two re-scorings for 15 and 2 hours.

Licences and data terms. The code is under the MIT licence. Derived data keep the terms and the citation rules of their sources. RouteViews data are under CC BY 4.0, and Cloudflare Radar data under CC BY-NC 4.0.

PCH states a Creative Commons Attribution-ShareAlike licence for its routing data [3]. Its page names CC BY-SA 4.0 but links to CC BY-NC-SA 4.0 [2], and the ShareAlike terms of the two differ. Because the two licences conflict, the scoring bundle holds no PCH-derived file. The 16 October package predates this check. It holds only derived PCH counts and AS sets, in 2024_pchorigin and 2024_leakpch, which we read as under the linked CC BY-NC-SA 4.0. The permanent archive of the final version will omit these two sets, as the scoring bundle does.

The CAIDA datasets are under the CAIDA Acceptable Use Agreement, which requires citation, and GRIP permits academic and educational use only. BGPKIT states that its peer statistics are free to use for any legal purpose [1].

The RIPE NCC requires attribution for commercial use of RIS data [4]. Its terms of service forbid redistributing data of its public services unless the use permits it (Art. 3.5) [6]. We therefore release no RIS archive file. The event tables hold derived research results, event times and AS sets, and our data card names RIS as their source. We read these derived tables as outside the restriction, which in our reading covers the RIS data themselves. The RIPE NCC has not confirmed this reading.

We found no published terms for the RIPE NCC RPKI archive [5] or for RoVista [7] when we checked on 31 October 2026. The repository terms of the RIPE NCC allow scientific research with its RPKI objects. We did not check the terms of the other four trust anchors. From them, the package holds only derived states, counts and shares, not ROA files or per-AS RoVista ratios.

The package does not include raw archives in the Multi-Threaded Routing Toolkit (MRT) format, raw third-party catalogues, CAIDA relationship files or bulk graph tensors. To rebuild the graph tensors, replay the public archives and obtain the topology inputs.

Compute and storage. Ingestion streams 2.8 to 3.9 TB of raw archives per year without storing them. The pass over 2024 took about 11 hours on cluster CPU nodes and kept 16.3 GB of prefix events. The learned forecast models trained on Quadro RTX 8000 GPUs, at 0.42 to 0.61 GPU-hours per core fit on 3,216 training events. The slow-regime ladder ran on CPUs, and each evaluation run of R2 requested about 2 TB of memory on one node. The package holds 3.56 GB in 2,194 files, and its 140 code tests pass in 22 s on a CPU. A rebuild from raw archives repeats the ingestion and needs the CAIDA files.

S5 Supplementary tables

Figure S1 and Tables S3 and S4 give the return by year and by identifier gap. Tables S5, S6 and S9 give the month-boundary and leaker sensitivities of the AP contrasts. Table S7 gives the set errors and the absolute precision at ten, and Table S8 the leaker recurrence. Table S10 compares the issue-time slice with persistence, and Table S11 gives the audit of R1 test leaks. Table S12 gives the secondary lanes of R2 from the two disclosed re-scorings. Tables S4 to S11 hold revision analyses, which we designed after the R1 and R2 results.

References

[1] BGPKIT. 2026. BGPKIT: BGP data processing tools (bgpkit-parser, bgpkit-broker) and peer statistics. https://bgpkit.com/ and https://github.com/bgpkit. Retrieved October 29, 2026 from https://data.bgpkit.com/peer-stats/

[2] Creative Commons. 2013. Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0), Legal Code. Retrieved November 1, 2026 from https://creativecommons.org/licenses/by-nc-sa/4.0/legalcode

[3] Packet Clearing House. 2026. MRT Routing Updates. PCH route collector archive. Routing data under a Creative Commons Attribution-ShareAlike licence. Retrieved October 30, 2026 from https://downloads.pch.net/BGP_Routing_Data

[4] RIPE NCC. 2026. Routing Information Service (RIS) raw data. https://ris.ripe.net/. Retrieved September 20, 2026 from https://data.ris.ripe.net/

[5] RIPE NCC. 2026. RPKI Repository Archive. Daily ROA files for the five RIR trust anchors. Retrieved October 30, 2026 from https://ftp.ripe.net/rpki/

[6] RIPE NCC. 2026. Terms of Service. https://www.ripe.net/about-us/legal/terms-of-service/. Retrieved November 1, 2026 from https://www.ripe.net/about-us/legal/terms-of-service/

[7] RoVista. 2026. RoVista. Per-AS ROV ratio series from the public API, https://api.rovista.netsecurelab.org/rovista/api. Retrieved October 31, 2026 from https://rovista.netsecurelab.org/

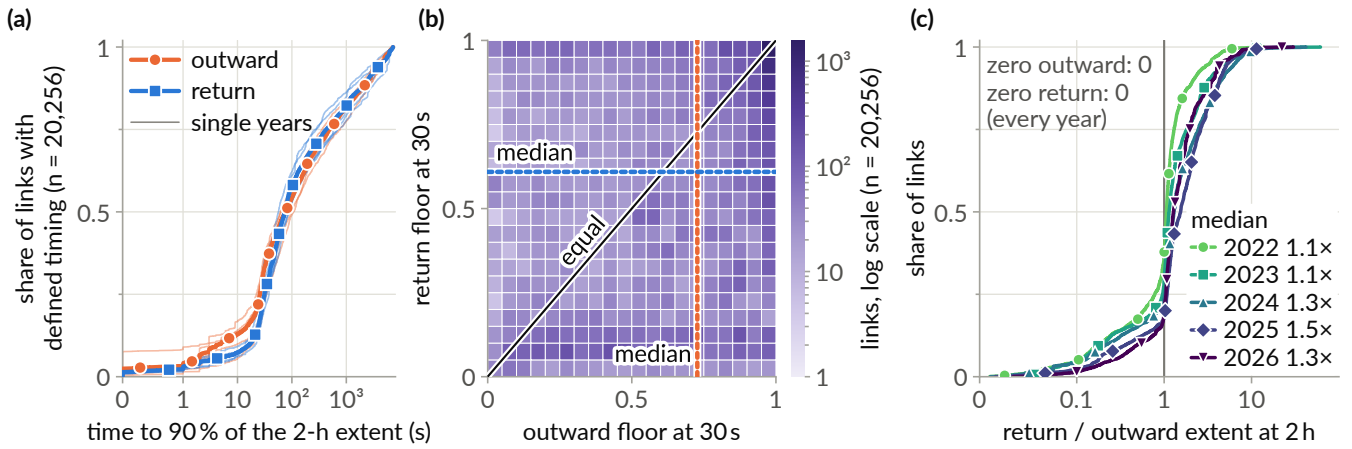


Figure S1: Matched outward and return transitions, 2022–2026. (a) Cumulative distributions of t_{90}, pooled over links, with thin lines for single years. (b) Per-link 30-s persistence floor, outward against return, with a logarithmic density scale, the marginal medians and the diagonal of equal floors. (c) Per-link ratio of return to outward extent by year. Links with undefined timing or ratios are excluded and counted. Tables 3 to S4 give the values by year and by identifier gap.

Table S3: Matched transitions by identifier gap and year. Share of links: of all links of that year. J_{30}: 30-s persistence floor of the outward (O) and return (R) footprint. Paired ΔJ: R minus O. Return slower: as in Table 3. Intervals: 95 % component bootstrap. Row labels: < 1 min is below 1 min, 1–15 min is at least 1 and below 15 min, and ≥ 15 min is at least 15 min and below 24 h.

gap	year	links	share of links	J_{30}, O / R	paired ΔJ [95 % CI]	t_{90} p50 (s) O / R	return slower	extent R/O, p50 [95 % CI]
< 1 min	2022	1,096	55 %	0.58 / 0.47	−0.104 [−0.12, −0.09]	100 / 114	65 %	1.09 [1.07, 1.10]
	2023	2,006	65 %	0.61 / 0.49	−0.119 [−0.13, −0.10]	77 / 109	73 %	1.18 [1.16, 1.19]
	2024	3,216	60 %	0.69 / 0.53	−0.152 [−0.16, −0.14]	59 / 93	75 %	1.36 [1.36, 1.39]
	2025	4,306	64 %	0.72 / 0.49	−0.227 [−0.24, −0.22]	44 / 68	67 %	1.57 [1.53, 1.65]
	2026	1,838	59 %	0.70 / 0.58	−0.114 [−0.13, −0.10]	59 / 74	71 %	1.32 [1.28, 1.35]
1–15 min	2022	279	14 %	0.55 / 0.64	+0.088 [+0.05, +0.13]	132 / 66	41 %	1.06 [1.04, 1.09]
	2023	414	13 %	0.56 / 0.58	+0.021 [−0.01, +0.05]	176 / 84	45 %	1.11 [1.09, 1.14]
	2024	880	16 %	0.58 / 0.60	+0.021 [−0.00, +0.05]	168 / 88	52 %	1.22 [1.19, 1.26]
	2025	1,486	22 %	0.64 / 0.52	−0.123 [−0.14, −0.10]	186 / 84	41 %	1.65 [1.60, 1.72]
	2026	479	15 %	0.61 / 0.63	+0.018 [−0.01, +0.05]	113 / 91	44 %	1.17 [1.14, 1.22]
≥ 15 min	2022	607	31 %	0.59 / 0.73	+0.132 [+0.10, +0.17]	105 / 48	36 %	0.95 [0.86, 1.00]
	2023	689	22 %	0.59 / 0.69	+0.100 [+0.07, +0.13]	186 / 56	37 %	1.04 [1.01, 1.07]
	2024	1,259	24 %	0.66 / 0.69	+0.034 [+0.01, +0.06]	80 / 46	39 %	1.14 [1.11, 1.15]
	2025	892	13 %	0.63 / 0.72	+0.092 [+0.07, +0.12]	122 / 42	36 %	1.13 [1.09, 1.15]
	2026	809	26 %	0.73 / 0.79	+0.062 [+0.04, +0.09]	55 / 30	41 %	1.26 [1.22, 1.40]

Table S4: Recovery intervals with dependence between links (revision analysis). Component: the published bootstrap over connected components of shared events. Day block: UTC days of the outward identifier as resampling units, with days joined by a shared event merged, 10,000 draws. *: the pooled extent ratio has no published interval, so this row uses the component bootstrap with 10,000 draws (seed 1701). Gap labels as in Table S3: ≥ 15 min is at least 15 min and below 24 h. Largest units: in 2025, one day holds 1,013 of the 4,306 links with a gap under 1 min. With a gap of at least 15 min, 116 of 1,259 links fall on 2024-10-17, with a mean ΔJ of −0.55, and 107 of 809 on 2026-08-01, with −0.18. Without these days, the means are +0.09 and +0.10, a post hoc check of these two years only.

	2022	2023	2024	2025	2026
day units (all links)	346	239	341	350	232
max. links per unit	36	75	342	1,017	242
all links					
paired ΔJ	−0.005	−0.052	−0.080	−0.161	−0.049
component CI	[−0.020, +0.011]	[−0.064, −0.040]	[−0.089, −0.070]	[−0.171, −0.152]	[−0.061, −0.037]
day-block CI	[−0.025, +0.016]	[−0.073, −0.030]	[−0.125, −0.039]	[−0.215, −0.094]	[−0.074, −0.022]
extent R/O, p50	1.06	1.14	1.29	1.54	1.28
component CI*	[1.05, 1.07]	[1.12, 1.16]	[1.28, 1.32]	[1.50, 1.56]	[1.25, 1.31]
day-block CI	[1.05, 1.08]	[1.10, 1.18]	[1.19, 1.45]	[1.20, 1.85]	[1.19, 1.45]
identifier gap < 1 min					
paired ΔJ	−0.104	−0.119	−0.152	−0.227	−0.114
component CI	[−0.122, −0.088]	[−0.134, −0.104]	[−0.162, −0.141]	[−0.237, −0.217]	[−0.128, −0.102]
day-block CI	[−0.126, −0.082]	[−0.147, −0.091]	[−0.196, −0.116]	[−0.277, −0.157]	[−0.149, −0.084]
extent R/O, p50	1.09	1.18	1.36	1.57	1.32
component CI	[1.07, 1.10]	[1.16, 1.19]	[1.36, 1.39]	[1.53, 1.65]	[1.28, 1.35]
day-block CI	[1.07, 1.11]	[1.12, 1.24]	[1.25, 1.83]	[1.20, 2.00]	[1.20, 1.50]
identifier gap 1–15 min					
paired ΔJ	+0.088	+0.021	+0.021	−0.123	+0.018
component CI	[+0.049, +0.129]	[−0.008, +0.052]	[−0.003, +0.047]	[−0.144, −0.103]	[−0.014, +0.049]
day-block CI	[+0.042, +0.135]	[−0.013, +0.055]	[−0.061, +0.114]	[−0.222, +0.019]	[−0.043, +0.086]
extent R/O, p50	1.06	1.11	1.22	1.65	1.17
component CI	[1.04, 1.09]	[1.09, 1.14]	[1.19, 1.26]	[1.60, 1.72]	[1.14, 1.22]
day-block CI	[1.02, 1.10]	[1.08, 1.17]	[1.12, 1.82]	[1.17, 2.00]	[1.12, 1.39]
identifier gap ≥ 15 min					
paired ΔJ	+0.132	+0.100	+0.034	+0.092	+0.062
component CI	[+0.098, +0.165]	[+0.071, +0.129]	[+0.011, +0.058]	[+0.065, +0.119]	[+0.036, +0.087]
day-block CI	[+0.093, +0.171]	[+0.060, +0.141]	[−0.077, +0.125]	[+0.035, +0.145]	[−0.025, +0.165]
extent R/O, p50	0.95	1.04	1.14	1.13	1.26
component CI	[0.86, 1.00]	[1.01, 1.07]	[1.11, 1.15]	[1.09, 1.15]	[1.22, 1.40]
day-block CI	[0.85, 1.01]	[1.00, 1.09]	[1.05, 1.26]	[1.07, 1.18]	[1.08, 1.80]

Table S5: Month-boundary sensitivity of the AP contrasts (revision analysis). Paired difference over eligible events, mean over five training seeds, with ordinary 95 % intervals from resampling training seeds and UTC-day blocks, each incident in the day of its first onset (10,000 repetitions, seed 1701). *All*: the published values. *No d1*: events whose identifier date is the first of a month removed; *no d1–3*: first to third removed. *Native*: SAGE with history minus native history (online history on its native candidates). The intervals are unadjusted and descriptive. Last rows: eligible events.

	R1 (Sep–Dec 2025)			R2 (1 Sep–16 Oct 2026)		
Contrast	All	No d1	No d1–3	All	No d1	No d1–3
C1@30 s	+0.0183 [+0.0115, +0.0252]	+0.0182 [+0.0117, +0.0246]	+0.0187 [+0.0114, +0.0261]	+0.0246 [+0.0172, +0.0329]	+0.0246 [+0.0171, +0.0330]	+0.0217 [+0.0154, +0.0285]
C2@30 s	+0.0054 [−0.0003, +0.0107]	+0.0066 [+0.0008, +0.0118]	+0.0068 [+0.0005, +0.0124]	+0.0003 [−0.0075, +0.0079]	+0.0013 [−0.0065, +0.0091]	−0.0010 [−0.0080, +0.0058]
C3@30 s	+0.0137 [+0.0079, +0.0180]	+0.0130 [+0.0069, +0.0178]	+0.0130 [+0.0066, +0.0180]	+0.0267 [+0.0179, +0.0361]	+0.0248 [+0.0156, +0.0345]	+0.0215 [+0.0137, +0.0292]
C4@30 s	+0.0633 [+0.0570, +0.0697]	+0.0629 [+0.0571, +0.0686]	+0.0640 [+0.0572, +0.0706]	+0.0660 [+0.0596, +0.0731]	+0.0622 [+0.0568, +0.0675]	+0.0614 [+0.0557, +0.0672]
Native@30 s	−0.0237 [−0.0307, −0.0172]	−0.0199 [−0.0264, −0.0140]	−0.0200 [−0.0270, −0.0136]	−0.0298 [−0.0406, −0.0197]	−0.0251 [−0.0344, −0.0161]	−0.0273 [−0.0363, −0.0186]
Eligible events, 30 s	2,661	2,229	1,888	1,004	860	715
C1@60 s	+0.0169 [+0.0118, +0.0215]	+0.0169 [+0.0119, +0.0213]	+0.0176 [+0.0120, +0.0224]	+0.0145 [+0.0069, +0.0227]	+0.0159 [+0.0076, +0.0247]	+0.0132 [+0.0050, +0.0223]
C2@60 s	+0.0022 [−0.0033, +0.0078]	+0.0037 [−0.0019, +0.0091]	+0.0039 [−0.0021, +0.0098]	−0.0042 [−0.0113, +0.0033]	−0.0012 [−0.0089, +0.0063]	−0.0044 [−0.0122, +0.0028]
C3@60 s	+0.0126 [+0.0067, +0.0187]	+0.0117 [+0.0058, +0.0182]	+0.0124 [+0.0063, +0.0191]	+0.0203 [+0.0138, +0.0264]	+0.0186 [+0.0117, +0.0253]	+0.0169 [+0.0100, +0.0235]
C4@60 s	+0.0591 [+0.0532, +0.0650]	+0.0572 [+0.0519, +0.0624]	+0.0579 [+0.0521, +0.0636]	+0.0645 [+0.0565, +0.0725]	+0.0590 [+0.0519, +0.0662]	+0.0594 [+0.0513, +0.0674]
Native@60 s	−0.0243 [−0.0317, −0.0172]	−0.0194 [−0.0259, −0.0132]	−0.0194 [−0.0264, −0.0125]	−0.0287 [−0.0391, −0.0180]	−0.0215 [−0.0306, −0.0130]	−0.0248 [−0.0340, −0.0160]
Eligible events, 60 s	2,130	1,775	1,516	842	708	585

Table S6: Leaker dependence of the C2 and native AP contrasts (revision analysis). Paired AP difference over eligible events, mean over five training seeds, as published. Ordinary 95 % intervals resample training seeds and then one of three units: UTC-day blocks with each incident in the day of its first onset (the published intervals), whole leaker ASes, or leakers and days drawn independently, with each event weighted by the product of the multiplicities of its leaker and its day (leaker × day). The leaker is the AS that leaks the route, read from the event identifier. 10,000 repetitions, seed 1701. Leakers: distinct leaker ASes among the eligible events, with the share of events of the largest leaker. Native: SAGE with history minus online history on its native candidates. Descriptive, no test family.

Period	Contrast	Diff.	Day blocks (published)	Leaker blocks	Leaker × day	Leakers (largest)
R1	C2@30 s	+0.0054	[−0.0003, +0.0107]	[−0.0002, +0.0108]	[−0.0014, +0.0125]	1,226 (1.1 %)
	Native@30 s	−0.0237	[−0.0307, −0.0172]	[−0.0302, −0.0175]	[−0.0327, −0.0146]	1,226 (1.1 %)
	C2@60 s	+0.0022	[−0.0033, +0.0078]	[−0.0033, +0.0078]	[−0.0059, +0.0104]	1,085 (1.2 %)
	Native@60 s	−0.0243	[−0.0317, −0.0172]	[−0.0311, −0.0177]	[−0.0350, −0.0139]	1,085 (1.2 %)
R2	C2@30 s	+0.0003	[−0.0075, +0.0079]	[−0.0075, +0.0079]	[−0.0099, +0.0111]	666 (1.1 %)
	Native@30 s	−0.0298	[−0.0406, −0.0197]	[−0.0388, −0.0210]	[−0.0437, −0.0157]	666 (1.1 %)
	C2@60 s	−0.0042	[−0.0113, +0.0033]	[−0.0108, +0.0027]	[−0.0146, +0.0070]	580 (1.0 %)
	Native@60 s	−0.0287	[−0.0391, −0.0180]	[−0.0368, −0.0204]	[−0.0437, −0.0141]	580 (1.0 %)

Table S7: Set errors and absolute precision at ten of the R1 and R2 forecasts (revision analysis, descriptive, no test family). A: ASes added to the seeds, R1 only, because the R2 records hold no forecast sets. Added: mean number of added ASes per event. Adds none: share of events whose forecast equals the seed set. B: P@10, the share of new members among the first ten ranked candidates, over eligible events. For R1, panel B does not repeat the J, Any and Mean values of the arms in panel A (–). $J-$ persistence: paired difference with an ordinary 95 % interval over training seeds and day blocks (10,000 repetitions, seed 1701), unadjusted. In R2, the Jaccard of persistence is $s/(s+R)$ for s seeds and R new members, and a forecast with Jaccard J adds $s(1/J-1)$ ASes to an event without new members. In R1, both identities hold on every record. Both panels: no-growth events have no new members, and every addition on them is false. Any: share of them with an addition. Mean: additions per such event. J: Jaccard against the full target, over all scored events. Learned arms: mean over five training seeds. Under each horizon: scored events, events without new members and, in B, eligible events. The frozen 2025 SAGE exists in R2 at 60 s only.

A. Added ASes

				No-growth events			
Period	Δ	Forecast	Added	Any	Mean	Adds none	J
R1	30 s	SAGE, history	0.34	0.006	0.02	0.946	0.702
	(3,543/882)	Online history (mean)	56.75	1.000	48.76	0.000	0.384
		Online history (median)	26.78	0.870	16.46	0.071	0.586
		Persistence	0.00	0.000	0.00	1.000	0.702
R1	60 s	SAGE, history	0.01	0.000	0.00	0.996	0.801
	(3,543/1,413)	Online history (mean)	42.77	0.983	36.89	0.010	0.477
		Online history (median)	16.17	0.631	9.31	0.265	0.728
		Persistence	0.00	0.000	0.00	1.000	0.801

B. Precision at ten, Jaccard and false additions

						No-growth events	
Period	Δ	Forecast	P@10	J	$J-$ persistence	Any	Mean
R1	30 s	SAGE, history	0.271	–	+0.000177 [+0.000061, +0.000301]	–	–
	(3,543/882/2,661)	Online history (mean)	0.270	–	−0.3186 [−0.3323, −0.3034]	–	–
		Online history (median)	0.270	–	−0.1160 [−0.1233, −0.1085]	–	–
		Online history, native	0.270	0.384	−0.3179 [−0.3317, −0.3027]	1.000	48.76
R1	60 s	SAGE, history	0.248	–	+0.0000083 [−0.0000001, +0.0000228]	–	–
	(3,543/1,413/2,130)	Online history (mean)	0.254	–	−0.3233 [−0.3359, −0.3096]	–	–
		Online history (median)	0.254	–	−0.0721 [−0.0778, −0.0663]	–	–
		Online history, native	0.254	0.477	−0.3231 [−0.3356, −0.3092]	0.983	36.89
R2	30 s	Persistence	–	0.671	–	0	0
	(1,289/285/1,004)	SAGE, history	0.271	0.671	−0.000210 [−0.000482, +0.000018]	0.011	0.03
		Online history (mean)	0.295	0.381	−0.2900 [−0.3154, −0.2627]	1.000	52.56
		Online history (median)	0.295	0.569	−0.1020 [−0.1158, −0.0884]	0.786	21.62
		Online history, native	0.291	0.382	−0.2898 [−0.3154, −0.2619]	1.000	52.56
R2	60 s	Persistence	–	0.776	–	0	0
	(1,289/447/842)	SAGE, history	0.241	0.776	−0.000082 [−0.000187, −0.000012]	0.003	0.01
		SAGE, frozen 2025	0.223	0.776	−0.000006 [−0.000023, +0.000006]	0.000	0.00
		Online history (mean)	0.260	0.484	−0.2919 [−0.3156, −0.2675]	0.982	37.29
		Online history (median)	0.260	0.706	−0.0693 [−0.0795, −0.0589]	0.575	10.61
		Online history, native	0.256	0.483	−0.2923 [−0.3163, −0.2676]	0.982	37.29

Table S8: Leaker recurrence (post hoc revision analysis, Sections S1.3 and S1.4). A: C2 and the native contrast by leaker class. A repeat leaker led at least one completed earlier event of the history library, and a first-seen leaker did not. Paired AP difference over the eligible events of the class, mean over five training seeds, with ordinary 95 % intervals over training seeds and then day blocks, or leakers and days drawn independently (leaker × day). Repeat − first: difference of the two class contrasts, from one joint draw. B: rankings of the native candidates of online history in R1, without training. Same leaker: same-leaker-only ranking, by completed earlier events of this leaker. Same leaker, then all: same-leaker-first ranking, ties broken by all completed earlier events. All rankings use the released seed sets, and *n* counts the eligible events that keep a new member under them. AP − SAGE compares complete systems on the exact-input events, a selected subset. 10,000 repetitions, unadjusted, no test family.

A. C2 and native contrasts by leaker class

Period	Contrast	Class	*n* (leakers)	Diff.	Day blocks	Leaker × day
R1	C2@30 s	First seen	420 (413)	+0.0131	[+0.0058, +0.0204]	[+0.0021, +0.0247]
		Repeat	2,241 (924)	+0.0040	[−0.0018, +0.0094]	[−0.0033, +0.0115]
		Repeat − first		−0.0090	[−0.0153, −0.0029]	[−0.0208, +0.0021]
	Native@30 s	First seen	420 (413)	−0.0112	[−0.0207, −0.0021]	[−0.0267, +0.0038]
		Repeat	2,241 (924)	−0.0260	[−0.0333, −0.0192]	[−0.0358, −0.0164]
		Repeat − first		−0.0148	[−0.0236, −0.0062]	[−0.0309, +0.0009]
	C2@60 s	First seen	335 (332)	+0.0109	[+0.0012, +0.0207]	[−0.0058, +0.0275]
		Repeat	1,795 (826)	+0.0006	[−0.0052, +0.0066]	[−0.0081, +0.0093]
		Repeat − first		−0.0102	[−0.0203, −0.0006]	[−0.0282, +0.0074]
	Native@60 s	First seen	335 (332)	−0.0091	[−0.0203, +0.0025]	[−0.0293, +0.0108]
		Repeat	1,795 (826)	−0.0272	[−0.0354, −0.0193]	[−0.0388, −0.0161]
		Repeat − first		−0.0181	[−0.0304, −0.0061]	[−0.0404, +0.0040]
R2	C2@30 s	First seen	124 (123)	+0.0087	[−0.0051, +0.0255]	[−0.0129, +0.0375]
		Repeat	880 (555)	−0.0009	[−0.0089, +0.0069]	[−0.0114, +0.0104]
		Repeat − first		−0.0096	[−0.0258, +0.0042]	[−0.0393, +0.0140]
	Native@30 s	First seen	124 (123)	−0.0199	[−0.0411, +0.0031]	[−0.0520, +0.0164]
		Repeat	880 (555)	−0.0312	[−0.0417, −0.0213]	[−0.0453, −0.0170]
		Repeat − first		−0.0113	[−0.0325, +0.0073]	[−0.0476, +0.0207]
	C2@60 s	First seen	107 (106)	+0.0011	[−0.0118, +0.0138]	[−0.0210, +0.0214]
		Repeat	735 (484)	−0.0050	[−0.0127, +0.0032]	[−0.0163, +0.0074]
		Repeat − first		−0.0061	[−0.0200, +0.0086]	[−0.0283, +0.0190]
	Native@60 s	First seen	107 (106)	−0.0193	[−0.0386, −0.0009]	[−0.0495, +0.0091]
		Repeat	735 (484)	−0.0300	[−0.0410, −0.0188]	[−0.0456, −0.0144]
		Repeat − first		−0.0107	[−0.0298, +0.0084]	[−0.0417, +0.0222]

B. R1, history rankings on the native candidates

Δ	Ranking	Class	*n* (exact)	AP / P@10	AP − online history	AP − SAGE (exact)
30 s	Online history	All	2,635 (1,866)	0.234 / 0.268	–	+0.0214 [+0.0144, +0.0288]
		First seen	418 (279)	0.219 / 0.242	–	+0.0103 [−0.0005, +0.0208]
		Repeat	2,217 (1,587)	0.236 / 0.273	–	+0.0233 [+0.0158, +0.0313]
	Same leaker	All	2,635 (1,866)	0.236 / 0.261	+0.0019 [−0.0115, +0.0140]	+0.0463 [+0.0327, +0.0587]
		First seen	418 (279)	0.014 / 0.012	−0.2044 [−0.2277, −0.1821]	−0.1684 [−0.1929, −0.1454]
		Repeat	2,217 (1,587)	0.277 / 0.307	+0.0408 [+0.0277, +0.0530]	+0.0840 [+0.0705, +0.0973]
	Same leaker, then all	All	2,635 (1,866)	0.307 / 0.315	+0.0729 [+0.0643, +0.0812]	+0.1043 [+0.0933, +0.1151]
		First seen	418 (279)	0.219 / 0.242	0 (same ranking)	+0.0103 [−0.0005, +0.0208]
		Repeat	2,217 (1,587)	0.323 / 0.329	+0.0867 [+0.0766, +0.0964]	+0.1208 [+0.1084, +0.1330]
60 s	Online history	All	2,109 (1,814)	0.225 / 0.254	–	+0.0229 [+0.0147, +0.0314]
		First seen	331 (278)	0.205 / 0.211	–	+0.0093 [−0.0036, +0.0223]
		Repeat	1,778 (1,536)	0.229 / 0.262	–	+0.0254 [+0.0166, +0.0346]
	Same leaker	All	2,109 (1,814)	0.216 / 0.236	−0.0094 [−0.0238, +0.0036]	+0.0225 [+0.0083, +0.0366]
		First seen	331 (278)	0.013 / 0.011	−0.1914 [−0.2158, −0.1681]	−0.1735 [−0.1960, −0.1525]
		Repeat	1,778 (1,536)	0.254 / 0.278	+0.0245 [+0.0096, +0.0388]	+0.0580 [+0.0437, +0.0726]
	Same leaker, then all	All	2,109 (1,814)	0.286 / 0.287	+0.0610 [+0.0516, +0.0706]	+0.0882 [+0.0765, +0.1002]
		First seen	331 (278)	0.205 / 0.211	0 (same ranking)	+0.0093 [−0.0036, +0.0223]
		Repeat	1,778 (1,536)	0.301 / 0.301	+0.0724 [+0.0612, +0.0840]	+0.1025 [+0.0897, +0.1157]

Table S9: Month-boundary subsets under leaker resampling (revision analysis). Paired AP difference over the kept eligible events, mean over five training seeds, with ordinary 95 % intervals over training seeds and then one of two units: whole leaker ASes, or leakers and days drawn independently (leaker × day). Table S5 gives the day-block intervals of the same subsets. 10,000 repetitions, seed 1701, unadjusted. No d1: events whose identifier date is the first of a month removed; no d1–3: first to third removed. Native: SAGE with history minus online history on its native candidates. Descriptive, no test family.

Period	Contrast	Subset	n	Diff.	Leaker blocks	Leaker × day
R1	C2@30 s	No d1	2,229	+0.0066	[+0.0007, +0.0120]	[−0.0005, +0.0136]
		No d1–3	1,888	+0.0068	[+0.0004, +0.0127]	[−0.0011, +0.0147]
	Native@30 s	No d1	2,229	−0.0199	[−0.0266, −0.0136]	[−0.0285, −0.0113]
		No d1–3	1,888	−0.0200	[−0.0272, −0.0131]	[−0.0297, −0.0105]
	C2@60 s	No d1	1,775	+0.0037	[−0.0018, +0.0095]	[−0.0042, +0.0119]
		No d1–3	1,516	+0.0039	[−0.0021, +0.0100]	[−0.0050, +0.0127]
	Native@60 s	No d1	1,775	−0.0194	[−0.0259, −0.0126]	[−0.0294, −0.0093]
		No d1–3	1,516	−0.0194	[−0.0265, −0.0121]	[−0.0303, −0.0087]
R2	C2@30 s	No d1	860	+0.0013	[−0.0070, +0.0093]	[−0.0092, +0.0128]
		No d1–3	715	−0.0010	[−0.0082, +0.0061]	[−0.0112, +0.0095]
	Native@30 s	No d1	860	−0.0251	[−0.0349, −0.0158]	[−0.0389, −0.0114]
		No d1–3	715	−0.0273	[−0.0365, −0.0185]	[−0.0415, −0.0139]
	C2@60 s	No d1	708	−0.0012	[−0.0088, +0.0062]	[−0.0128, +0.0105]
		No d1–3	585	−0.0044	[−0.0121, +0.0032]	[−0.0165, +0.0076]
	Native@60 s	No d1	708	−0.0215	[−0.0306, −0.0126]	[−0.0358, −0.0078]
		No d1–3	585	−0.0248	[−0.0346, −0.0153]	[−0.0405, −0.0101]

Table S10: Set forecasts against persistence on the prespecified issue-time slice (revision analysis). The slice and its threshold are as in Table 7: early growth of at least 12 at 30 s or 22 at 60 s, slice Jaccard with the threshold calibrated on validation slice events. Persistence forecasts the seed set, with the Jaccard $s/(s+R)$ in R2. J: slice Jaccard, mean over five training seeds for the learned arms. $J-$ persistence: paired difference over the slice events, with an ordinary 95 % interval over training seeds and then day blocks, 10,000 repetitions, seed 1701, unadjusted. The slice is prespecified, but this contrast is not. A row whose values are all below 0.001 in magnitude prints more decimals. The published slice C2 is SAGE with history minus online history, so it is about the distance of online history below persistence.

Period	Δ	Slice events	Forecast	J	$J-$ persistence
R1	30 s	966	Persistence	0.670	–
			SAGE, history	0.670	+0.0008 [+0.0003, +0.0014]
			Online history (mean)	0.573	−0.0968 [−0.1116, −0.0816]
R1	60 s	502	Persistence	0.729	–
			SAGE, history	0.729	+0.000039 [−0.000092, +0.000170]
			Online history (mean)	0.656	−0.0733 [−0.0903, −0.0559]
R2	30 s	372	Persistence	0.652	–
			SAGE, history	0.651	−0.0008 [−0.0028, +0.0007]
			Online history (mean)	0.557	−0.0953 [−0.1180, −0.0745]
R2	60 s	211	Persistence	0.714	–
			SAGE, history	0.714	−0.000012 [−0.000139, +0.000084]
			Online history (mean)	0.630	−0.0837 [−0.1083, −0.0596]
			SAGE, frozen 2025	0.714	−0.000059 [−0.000293, +0.000115]

Table S11: Audit of R1 test leaks (revision analysis). A script applies fixed rules, without human labels or arm results. It tests consistency with CAIDA relationships, not ground truth, so it does not measure the precision of the identifier. A: 30 events per stratum, drawn without replacement (seed 1701). Radar: matched to a Radar leak label. Volume: selected by message volume. Day 1: the identifier date is the first day of a month. Section S1.6 defines the classes. Counts with shares and Wilson 95 % intervals. Cohort: shares weighted by stratum size, with the same weights on the stratum Wilson limits (conservative, descriptive, not a joint interval). B: C2 and the native contrast (SAGE with history minus native history) in AP at 30 s over the eligible events of each population, with ordinary 95 % intervals over training seeds and day blocks (10,000 repetitions, seed 1701), unadjusted and descriptive over nine populations. The day-1 strata have only 6 and 7 day blocks. Only sampled events carry a class, so the row without the sampled changed or missing events removes only some such events. C: the same sample by evidence other than the identifier's relationship test (Section S1.6). The Radar strata carry 97 % of the cohort weight, and a Radar match selected their events, so 50 of their 54 supports come from the same labels. Path exploration can also make a short new route.

A: stratum	Events	Audited	Relationship-consistent	Relationship changed or missing	Ambiguous
Radar, day 1	476	30	29 (97 %; 83–99)	0 (0 %; 0–11)	1 (3 %; 1–17)
Radar, other days	2,968	30	29 (97 %; 83–99)	0 (0 %; 0–11)	1 (3 %; 1–17)
Volume, day 1	65	30	17 (57 %; 39–73)	2 (7 %; 2–21)	11 (37 %; 22–54)
Volume, other days	34	30	22 (73 %; 56–86)	8 (27 %; 14–44)	0 (0 %; 0–11)
Cohort, weighted	3,543		95.7 % (82–99)	0.4 % (0–12)	3.9 % (1–17)

B: population (AP, 30 s)	Eligible	C2	Native
All (published)	2,661	+0.0054 [−0.0003, +0.0107]	−0.0237 [−0.0307, −0.0172]
Without the sampled changed or missing	2,651	+0.0055 [−0.0002, +0.0107]	−0.0234 [−0.0304, −0.0170]
Radar, day 1	367	+0.0009 [−0.0101, +0.0171]	−0.0241 [−0.0354, −0.0062]
Radar, other days	2,195	+0.0064 [+0.0007, +0.0116]	−0.0195 [−0.0259, −0.0136]
Volume, day 1	65	−0.0085 [−0.0188, +0.0128]	−0.1506 [−0.1779, −0.1042]
Volume, other days	34	+0.0192 [−0.0102, +0.0460]	−0.0462 [−0.0921, −0.0024]
Audited sample	107	+0.0077 [−0.0074, +0.0277]	−0.0623 [−0.1003, −0.0194]
Audited, relationship-consistent	84	+0.0138 [−0.0038, +0.0360]	−0.0422 [−0.0804, −0.0063]
Audited, independently supported	61	+0.0251 [+0.0049, +0.0562]	−0.0031 [−0.0304, +0.0301]

C: stratum	Events	Audited	Independently supported	Independently contradicted	Uncertain
Radar, day 1	476	30	29 (97 %; 83–99)	0 (0 %; 0–11)	1 (3 %; 1–17)
Radar, other days	2,968	30	25 (83 %; 66–93)	0 (0 %; 0–11)	5 (17 %; 7–34)
Volume, day 1	65	30	1 (3 %; 1–17)	4 (13 %; 5–30)	25 (83 %; 66–93)
Volume, other days	34	30	16 (53 %; 36–70)	0 (0 %; 0–11)	14 (47 %; 30–64)
Cohort, weighted	3,543		83.4 % (67–92)	0.2 % (0–12)	16.4 % (8–33)

Table S12: Secondary lanes of the two disclosed re-scorings (R2). A role is one retained model configuration, horizon and seed. The 344 secondary roles and the 56 primary roles make up the 400 declared roles of R2. The five named lanes hold 224 roles, all scored. The four other groups hold 120 roles, among them 54 retained failures. The recorded paired contrasts shown here use the full-target Jaccard. H1 also records a two-threshold Jaccard. The per-event rows also hold AP and other metrics, which this table does not show. On this metric the learned set forecasts mostly equal persistence, so the lane differences reflect the inputs more than forecast skill. H1, H3 and the history lane compare each role with the matching updated model with history (same message passing, horizon and seed), or with the online history baseline, on the same events. H3 feeds seven collectors (RIS only, or mixed) instead of ten to the same weights or to a refitted model and scores against the fixed ten-collector target, so it does not measure global visibility. History withholding drops the past events that a source-unavailability rule marks, without retraining. The timing lane compares each horizon with 30 s on one fixed cohort, and later issue times observe more of the target. The 5, 10, 300 and 900 s arms are separate fits, while the 30 and 60 s arms reuse the core models, trained on the broader core population, so these differences do not isolate the issue time. In H3 and the timing lane, the arm without message passing moves by about as much as SAGE. The second re-scoring scored the stable-peer lane and excludes events whose stable-peer target and forecast are both empty (Section S3.4). No lane records an interval or a test.

Lane	Roles	Scored	Not scored	Recorded paired difference in full-target J (range over five seeds)
Candidate expansion (H1)	20	20	0	SAGE −0.0006 to +0.0003 (30 s), −0.0002 to +0.0002 (60 s); no message passing $\|\Delta\| < 0.0001$; 1,289 events
Seven of ten collectors (H3)	80	80	0	SAGE, 30 s / 60 s: RIS only, same weights −0.079 to −0.078 / −0.093; RIS only, refitted −0.074 to −0.073 / −0.087 to −0.084; mixed, same weights −0.104 to −0.103 / −0.122; mixed, refitted −0.097 to −0.094 / −0.116 to −0.111; 1,289 events
Stable peers	24	24	0	no contrast recorded; 1,286 (30 s) and 1,286 (60 s) events per role; 3 and 3 events with an empty stable-peer target and an empty forecast are not scorable (separate re-scoring)
Input timing	60	60	0	SAGE minus the same seed at 30 s: 5 s −0.213 to −0.212, 10 s −0.168 to −0.166, 60 s +0.099, 300 s +0.215, 900 s +0.251; 1,124 events
Origin changes	40	40	0	no contrast recorded; RPKI-invalid 14,004, valid 135,088 events per role
History delay, withholding	48	48	0	all 48 contrasts $\|\Delta\| < 0.0001$; 1,289 events
Ordinary temporal models	32	12	20	no contrast declared; TGN and mean aggregation (20 roles) are retained R1 failures
Count forecasts	34	0	34	not estimable: retained count and Chronos failures
Additional baselines	6	6	0	no contrast recorded; 1,289 events per role
Total	344	290	54	